Assignment
Subject: programing_fundamental_lab_4
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**GHAZI UNIVERSITY DERA GHAZI KHAN**

# Task1:

Write a program to print positive number entered by the user. If the user enters negative number print number entered is positive otherwise print number is negative.

```
#include<iostream>
using namespace std;
int main()
{
    int a;
    cout<<"enter a number: ";
    cin>>a;
    if(a>0)
    {
        cout<<"entered number is positive: "<<a;
    }
    else
    {
        cout<<"entered number is negative: "<<a;
    }
    return 0;
}
```

Output:

```
C:\Users\dell\Documents\lab4 task1.exe

enter a number: 8
entered number is positive: 8
--------------------------------
Process exited after 2.1 seconds with return value 0
Press any key to continue . . .
```

```
C:\Users\dell\Documents\lab4 task1.exe

enter a number: -9
entered number is negative: -9
--------------------------------
Process exited after 2.1 seconds with return value 0
Press any key to continue . . .
```

# Task2:

## Program to check whether an integer is positive, negative or zero.

```
#include<iostream>
using namespace std;
int main()
{
    int a;
    cout<<"enter a number: ";
    cin>>a;
    if(a>0)
    {
        cout<<"the entered number is positive: "<<a;
    }
    else if(a<0)
    {
        cout<<"entered number is negative: "<<a;
    }
    else
    {
        cout<<"entered number is ZERO: "<<a;
    }
    return 0;
}
```

## Output:

```
C:\Users\dell\Documents\lab4 task2.exe

enter a number: 12
the entered number is positive: 12
--------------------------------
Process exited after 2.1 seconds with return value 0
Press any key to continue . . .
```

```
C:\Users\dell\Documents\lab4 task2.exe

enter a number: -7
entered number is negative: -7
--------------------------------
Process exited after 2.1 seconds with return value 0
Press any key to continue . . .
```

```
C:\Users\dell\Documents\lab4 task2.exe

enter a number: 0
entered number is ZERO: 0
--------------------------------
Process exited after 2.1 seconds with return value 0
Press any key to continue . . .
```

# Task3:

Input: Mark. Process: If mark greater than and equal to 75, score will be A. If mark less than 75 and greater than and equal to 60, score will be B. If mark less than 60 and greater than and equal to 45, score will be C. If mark less than 45 and greater than 30, score will be D. Output: Print the grade of your score.

```
#include<iostream>
using namespace std;
int main()
{
    int marks;
    cout<<"enter your marks: ";
    cin>>marks;
    if(marks>=75)
    {
        cout<<"Grade : A";
    }
    else
    {
        if(marks>=60)
        {
            cout<<"Grade : B";
        }
        else
        {
            if(marks>=45)
            {
                cout<<"Grade : C";
            }
            else
            {
                if(marks>30)
                {
                    cout<<"Grade : D";
                }
                else
                {
                    cout<<"Better luck next time";
                }
            }
        }
    }
    return 0;
}
```

## Output:

```
C:\Users\dell\Documents\lab4 task3.exe

enter your marks: 91
Grade : A
--------------------------------
Process exited after 2.1 seconds with return value 0
Press any key to continue . . .
```

```
C:\Users\dell\Documents\lab4 task3.exe

enter your marks: 63
Grade : B
--------------------------------
Process exited after 2.1 seconds with return value 0
Press any key to continue . . .
```

```
C:\Users\dell\Documents\lab4 task3.exe

enter your marks: 50
Grade : C
--------------------------------
Process exited after 2.1 seconds with return value 0
Press any key to continue . . .
```

```
C:\Users\dell\Documents\lab4 task3.exe

enter your marks: 38
Grade : D
--------------------------------
Process exited after 2.1 seconds with return value 0
Press any key to continue . . .
```

```
C:\Users\dell\Documents\lab4 task3.exe

enter your marks: 22
Better luck next time
--------------------------------
Process exited after 2.1 seconds with return value 0
Press any key to continue . . .
```

# Task4:

Find Largest Number Using Nested if...else statement. Check whether the number entered by the user is positive or not. If it is positive then calculate how many digits the number have.

```
#include<iostream>
using namespace std;
int main()
{
    int a, n, b=0;
    cout<<"enter a number: ";
    cin>>a;
    if(a>0)
    {
        n=a;
        while(n>0)
        {
            n/=10;
            b++;
        }
        cout<<"the number is positive: "<<a<<endl;
        cout<<"number of digits: "<<b<<endl;
    }
    else
    {
        cout<<"entered number is not positive: "<<a;
    }
    return 0;
}
```

## Output:

```
C:\Users\dell\Documents\lab4 task4.exe

enter a number: 4821
the number is positive: 4821
number of digits: 4

--------------------------------
Process exited after 2.1 seconds with return value 0
Press any key to continue . . .
```

```
C:\Users\dell\Documents\lab4 task4.exe

enter a number: 7
the number is positive: 7
number of digits: 1

--------------------------------
Process exited after 2.1 seconds with return value 0
Press any key to continue . . .
```

```
C:\Users\dell\Documents\lab4 task4.exe
```

```
enter a number: -15
entered number is not positive: -15
--------------------------------
Process exited after 2.1 seconds with return value 0
Press any key to continue . . .
```